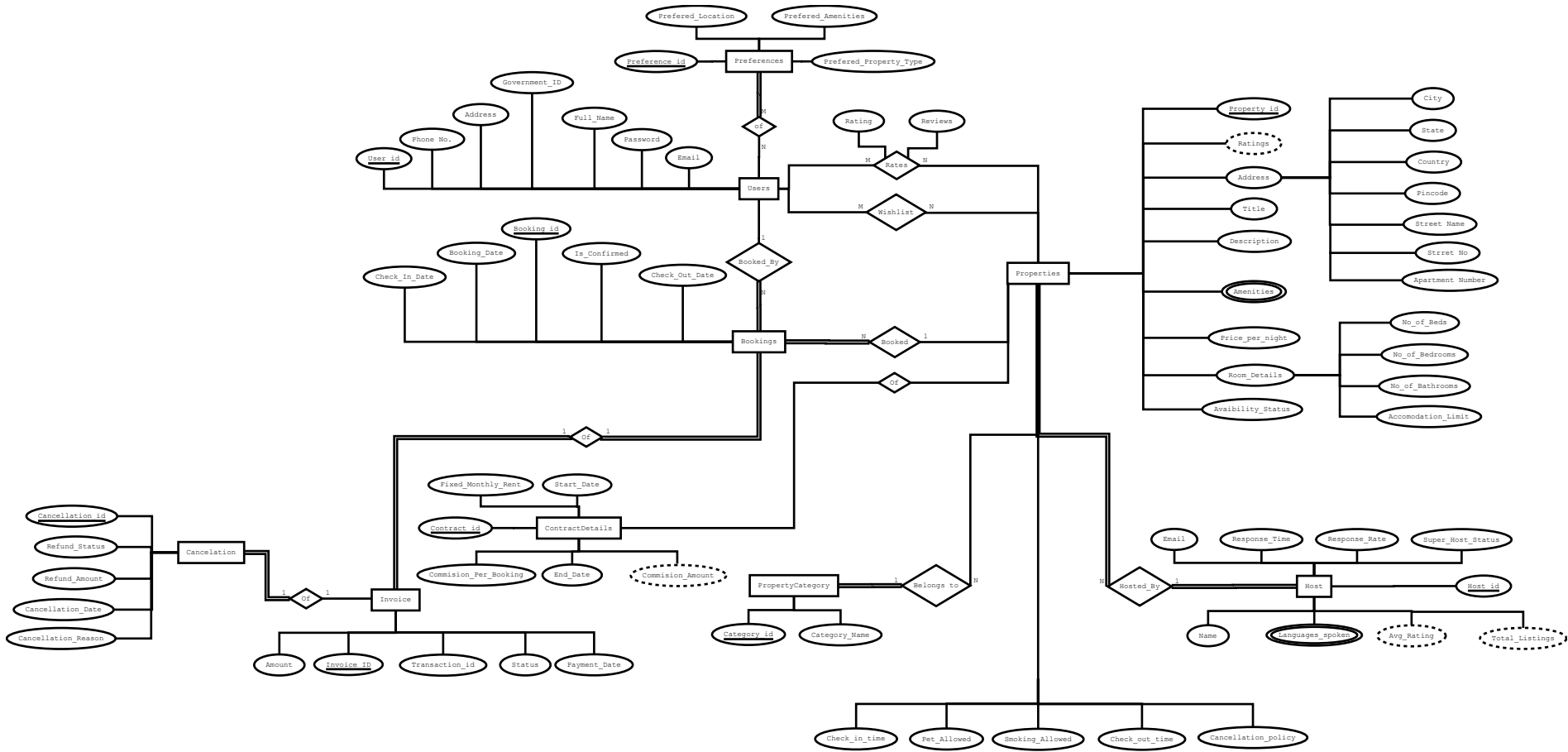
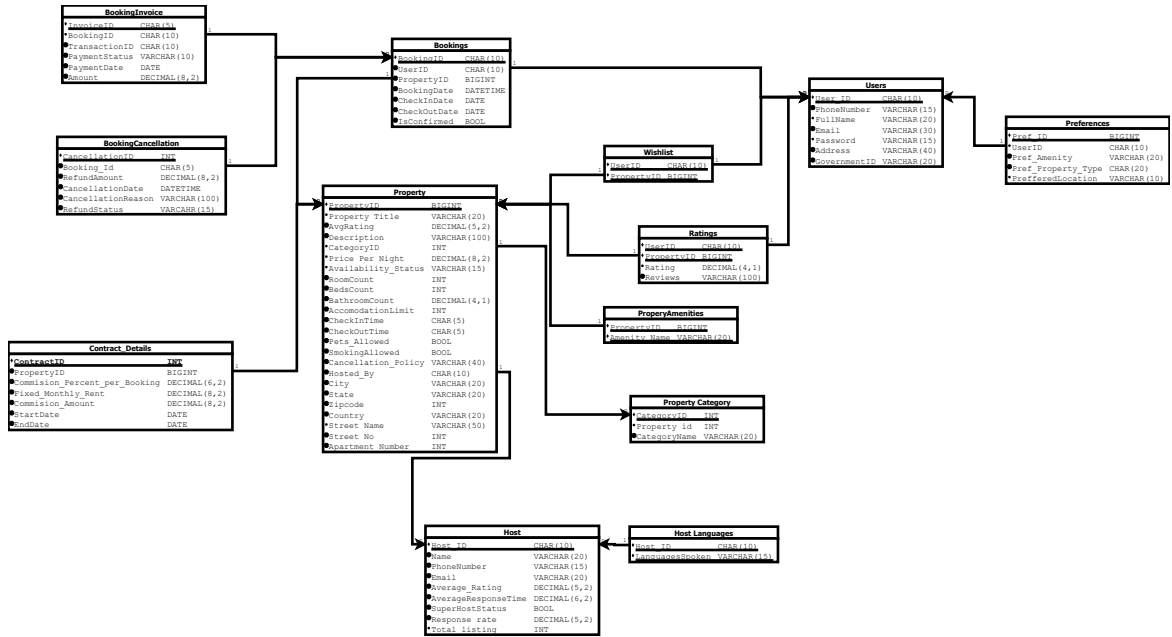






Normalisation Proofs

- Property Relation
- FD Set

PropertyID -> {Property Title, AvgRating, Description, CategoryID, Price Per Night, Availability_Status, RoomCount, BedsCount, BathroomCount, AccomodationLimit, CheckInTime, CheckOutTime, Pets_Allowed, SmokingAllowed, Cancellation_Policy, Hosted_By, City, State, Zipcode, Street name, Street Number, Apartment Num, Country}

Zipcode -> {City, State, Country}

- Min FD

PropertyID -> Property Title

PropertyID -> AvgRating

PropertyID -> Description

PropertyID -> CategoryID

PropertyID -> Price Per Night

PropertyID -> Availability_Status

PropertyID -> RoomCount

PropertyID -> BedsCount

PropertyID -> BathroomCount

PropertyID -> AccomodationLimit

PropertyID -> CheckInTime

PropertyID -> CheckOutTime

PropertyID -> Pets_Allowed

PropertyID -> SmokingAllowed

PropertyID -> Cancellation_Policy

PropertyID -> Hosted_By

PropertyID -> Zipcode

PropertyID -> Street Name

PropertyID -> Street Number

PropertyID -> Apartment Number

Zipcode -> City

Zipcode -> State

Zipcode -> Country

- Closure of PropertyID+ : {PropertyID,PropertyTitle,AvgRating,Description,CategoryID,Price Per Night,Availability_Status,RoomCount,BedsCount,BathroomCount,AccommodationLimit,CheckInTime,CheckOutTime,Pets_Allowed,SmokingAllowed,Cancellation_Policy,Hosted_By,City,Street Name,Street Number,,State,Zipcode,Country}
- Here key is PropertyID.
- The last three FDs ($Zipcode \rightarrow City$, $Zipcode \rightarrow State$, $Zipcode \rightarrow Country$) do not have PropertyID on the left-hand side. Therefore this relation is not in BCNF; it is in **2NF** because although every non-key attribute is fully functionally dependent on the key (no partial dependency), Zipcode transitively determines City, State, and Country. We can convert it to BCNF by decomposing into two relations:

- **R1** (PropertyID, PropertyTitle, AvgRating, Description, CategoryID, Price_Per_Night, Availability_Status, RoomCount, BedsCount, BathroomCount, AccommodationLimit, CheckInTime, CheckOutTime, Pets_Allowed, SmokingAllowed, Cancellation_Policy, Hosted_By, Zipcode, Street Name, Street Number, Apartment Number)

- **R2** (Zipcode, City, Country, State)

1. We did not decompose because it makes the database more practical and queries faster."

- **Contract_Details Relation**
- **FD set**

ContractID->{PropertyID,Commision_Percent_Per_Booking,Fixed_Monthly_Rent,
Commision_Amount,StartDate,EndDate}

- Min FD

ContractID->PropertyID

ContractID-> Commision_Percent_Per_Booking

ContractID-> Fixed_Monthly_Rent

ContractID-> Commision_Amount

ContractID-> StartDate

ContractID-> EndDate

- ClosureofContractID+ : { ContractIDPropertyID,Commision_Percent_Per_Booking,
Fixed_Monthly_Rent, Commision_Amount,StartDate,EndDate}

- Here key is ContractID.

- 'Contract_Details' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.

- **BookingCancellation Relation**
- **FD set**

CancellationID -> {BookingID,RefundAmount,CancellationDate,CancellationReason,RefundStatus}

- Min FD

CancellationID -> BookingID

CancellationID -> RefundAmount

CancellationID -> CancellationDate

CancellationID -> CancellationReason

CancellationID -> RefundStatus

- Closure of CancellationID+ : { CancellationID,BookingID,RefundAmount,CancellationDate,
CancellationReason,RefundStatus}

- Here key is CancellationID.

- 'BookingCancellation' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.

- **BookingInvoice Relation**
- **FD set**

InvoiceID -> {BookingID,TransactionID,PaymentStatus,PaymentDate,Amount}

- Min FD

InvoiceID -> BookingID

InvoiceID -> TransactionID

InvoiceID -> PaymentStatus

InvoiceID -> PaymentDate

InvoiceID -> Amount

- Closure of InvoiceID+ : {InvoiceID,BookingID,TransactionID,PaymentStatus,PaymentDate,Amount}

- Here key is InvoiceID

- 'BookingInvoice' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.
- Bookings Relation
- FD set

BookingID -> {UserID,PropertyID,BookingDate,CheckInDate,CheckOutDate,IsConfirmed}

- Min FD

BookingID -> UserID

BookingID -> PropertyID

BookingID -> BookingDate

BookingID -> CheckInDate

BookingID -> CheckOutDate

BookingID -> IsConfirmed

- ClosureofBookingID+ : {BookingIDUserID,PropertyID,BookingDate,CheckInDate,CheckOutDate,IsConfirmed}

- Here key is BookingID.

- 'Bookings' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.

- **Users Relation**
- **FD set**

User_ID -> {PhoneNumber,FullName,Email>Password,Address,GovernmentID}

- Min FD

User_ID -> PhoneNumber

User_ID -> FullName

User_ID -> Email

User_ID -> Password

User_ID -> Address

User_ID -> GovernmentID

- Closure of User_ID+ : {User_ID,PhoneNumber,FullName,Email>Password,Address,GovernmentID}
- Here key is User_ID.
- 'Users' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.
- **Preferences Relation**
- **FD set**

Pref_ID -> {UserID,Pref_Amenity,Pref_Property_Type,PrefferedLocation}

- Min FD

Pref_ID -> UserID

Pref_ID -> Pref_Amenity

Pref_ID -> Pref_Property_Type

Pref_ID -> PrefferedLocation

- Closure of Pref_ID+ : { Pref_ID ,UserID,Pref_Amenity,Pref_Property_Type,PrefferedLocation}
- Here key is Pref_ID.
- 'Preferences' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.
- **Ratings Relation**
- **FD set**

{User_ID,PropertyID} -> {Rating,Reviews}

- Min FD

{User_ID,PropertyID} -> Rating

{User_ID,PropertyID} -> Reviews

- Closure of {User_ID,PropertyID} + : {User_ID,PropertyID,Rating,Reviews}+

- Here key is {User_ID,PropertyID}

- 'Ratings' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.

- **Hosts Relation**
- **FD set**

- Min FD

Host_ID -> Name

Host_ID -> PhoneNumber

Host_ID -> Email

Host_ID -> Average_Rating

Host_ID -> AverageResponseTime

Host_ID -> SuperHostStatus

Host_ID -> Total Listing

Host_ID -> Response Rate

Closure of Host_ID+ : {Host_ID,Name,PhoneNumber,Email,Average_Rating,
AverageResponseTime,SuperHostStatus,Total_Listing,Response_Rate}

- Here key is Host_ID

- 'Hosts' relation is in BCNF because in minimal FD set all FDs have key on their left-hand side.

- **Wishlist Relation**
- Here key is {UserID,PropertyID}
- This relation is in BCNF because two attributes relation always in BCNF.

- **PropertyAmenities Relation**
- Here key is {PropertyID,Amenity_Name}
- This relation is in BCNF because two attributes relation always in BCNF.

- **Property Category Relation**
- **FD set**

CategoryID -> CategoryName

- Min FD

CategoryID -> CategoryName

Closure of CategoryID+ : {CategoryID, CategoryName}

- Here key is CategoryID.
- This relation is in BCNF because in minimal FD set FD have key on its left-hand side.
- Host Languages Relation
- Here key is {Host_ID , LanguagesSpeken}
- This relation is in BCNF because two attributes relation always in BCNF.